

Improving React Website Performance Using Lazy Loading and Image Optimization

Abstract

This study investigates methods for improving the speed and responsiveness of React websites. Many modern web applications suffer from slow loading times due to large images and unnecessary rendering processes. The research focuses on implementing lazy loading and image optimization techniques to improve performance. Results showed reduced page load times and smoother user experience.

Introduction

Modern websites rely heavily on animations, images, and interactive components. However, excessive assets can reduce performance and negatively affect user experience.

Problem Statement

Many React websites load slowly, especially on low-end devices and slower internet connections.

Objectives

Analyze causes of poor React performance, implement lazy loading, optimize images and assets, and measure performance improvements.

Conclusion

React website performance can be improved greatly using modern optimization techniques such as lazy loading and image compression.